

Two Pass Assembler

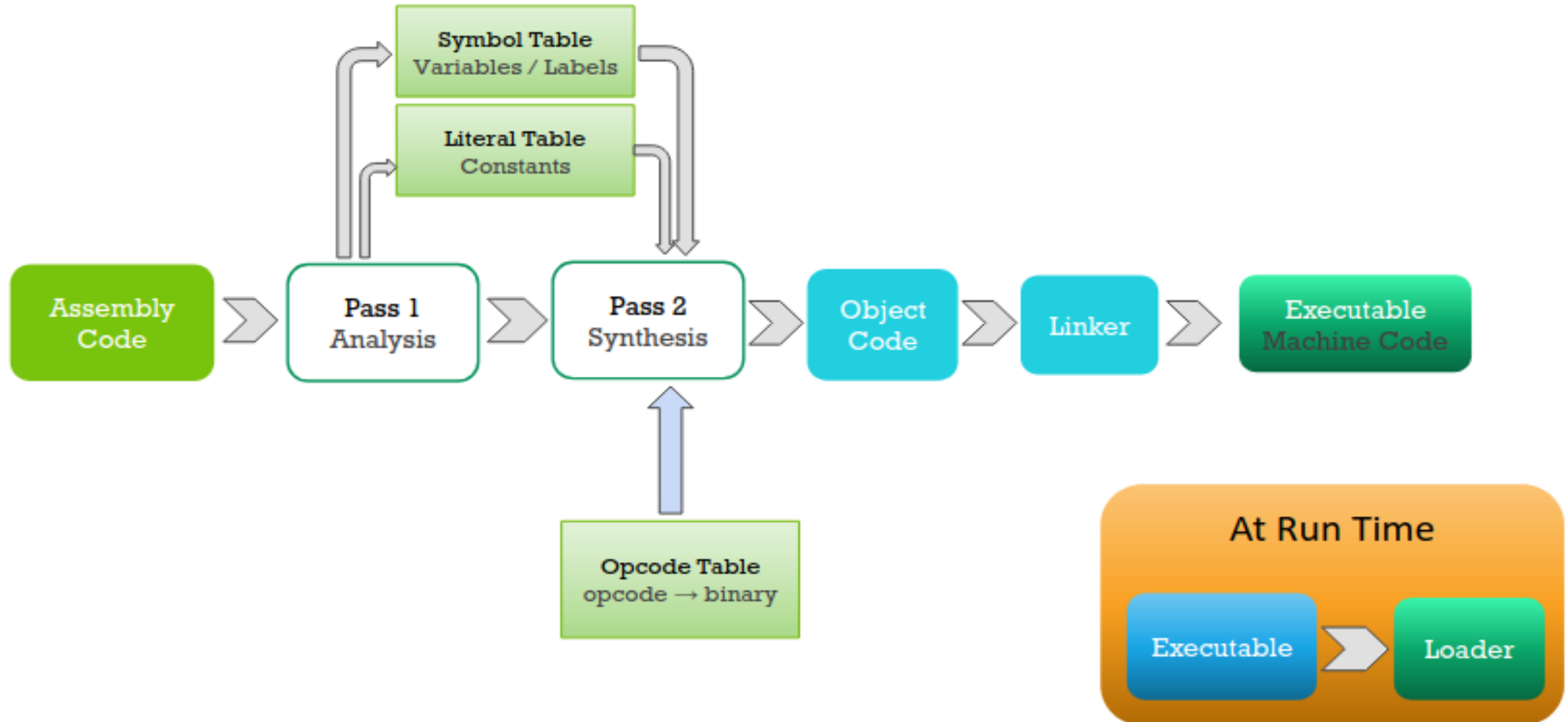
CS 348

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- Pass 1 – Analysis
- Intermediate Code Generation
- Pass 2 – Synthesis
- Output and system Integration

Two Pass Assembler Overview



Two Pass Assembler Overview

- Convert mnemonic operation codes to their machine language equivalents
- Convert symbolic operands to their equivalent machine addresses
- Build the machine instructions in the proper format
- Convert the data constants to internal machine representations
- Output the object program and assembly listing (optional).

Design of Two Pass Assembler

Pass I – Analysis

- Separate the symbols, Literals, Mnemonic and Operand fields
- Build the Symbol and Literal Table
- Perform the Location Counter processing
- Construct the intermediate representation

Pass II – Synthesis

- Synthesize the target code

Pass I – Analysis

Scan the program looking for symbols, labels, variables etc

- Initialize and update the Location Counter (LC)
- Identify and enter symbols into SYMTAB
- Collect literals into LITTAB
- Process Pseudo operations (macros / directives) through instruction classification
- Detect syntax and semantic errors

OUTPUT : Intermediate Code(IC)

Types of Assembly Language Statements

Three types of statements

- **Imperative (IS)**

Statements that translate into machine instruction to be performed during execution of the assembled program.

Ex. ADD, SUB, MOV, JMP

- **Declarative (DS)**

Statements that reserve areas of memory and associate names with them.

Construct memory word containing constants

Ex. EQU, DC

- **Assembly Directives (AD)**

Instruct the assembler to perform certain actions during assembly of a program

Ex. START 100, ORIGIN LABEL+5

Data Structures

- Opcode Table (OPTAB)
- Symbol Table (SYMTAB)
- Literal Table (LITTAB)
- Location Counter (LOCCTR)

OpCode Table (OPTAB)

Contain the opcodes and corresponding metadata

Fields

- Index
- Mnemonic
- Type
- OP-Code
- Length

Index	Mnemonic	TYPE	OP-Code	Length
1. Mnemonic Op-Code Table(MOT)				
1	MOVER	IS	01	01
2	MOVEM	IS	02	01
3	ADD	IS	03	01
4	SUB	IS	04	01
5	MULT	IS	05	01
6	DIV	IS	06	01
7	BC	IS	07	01
8	COMP	IS	08	01
9	PRINT	IS	09	01
10	READ	IS	10	01

OpCode Table (OPTAB)

OPTAB

Index	Mnemonic	TYPE	OP-Code	Length
1. Mnemonic Op-Code Table(MOT)				
1	MOVER	IS	01	01
2	MOVEM	IS	02	01
3	ADD	IS	03	01
4	SUB	IS	04	01
5	MULT	IS	05	01
6	DIV	IS	06	01
7	BC	IS	07	01
8	COMP	IS	08	01
9	PRINT	IS	09	01
10	READ	IS	10	01

Assembler Directive (AD) (POT)				
Index	Mnemonic	TYPE	OP-Code	Length
1	START	AD	01	-
2	END	AD	02	-
3	EQU	AD	03	-
4	ORIGIN	AD	04	-
5	LTORG	AD	05	-

Declaration Statements				
Index	Mnemonic	TYPE	OP-Code	Length
01	DS	DL	06	-
02	DC	DL	07	-

Symbol table (SYMTAB)

Table used for mapping symbolic names (labels) to addresses and values

Fields

- Sym_no
- Symbol
- Address

Sym_no	Symbol	Address
1	LABEL	304
2	LOOP	307

Literal Table (LITTAB)

Table used for mapping symbolic names (labels) to addresses and values

Fields

- Lit_no
- Literal
- Address

Lit_no	Literal	Address
1	= '8'	304
2	= '89'	307

Intermediate Code

Generated at the end of Pass I and used to synthesize object code in Pass II.

Why Intermediate Code is Needed

- Avoids re-parsing source code in Pass 2
- Encodes semantic meaning of each statement
- Simplifies code generation logic
- Makes assembler modular and clean

Format of Intermediate code Entry

(<Type>, <Code>) (<Operand1>) (<Operand2>)

Intermediate Code

Records:

- Type of statement (IS / DL / AD)
- Opcode or directive identifier
- Operand types (Register / Symbol / Literal / Constant)
- References to SYMTAB and LITTAB

Each line of the source program is classified as:

- **IS** – Imperative Statement (machine instruction)
- **DL** – Declarative Statement (storage / data)
- **AD** – Assembler Directive (control information)

Operand Representation

Operands are explicitly tagged:

- (R, r) Register operand
- (S, i) Symbol table index / symbol
- (L, i) Literal table index
- (C, v) Constant value

Facilitate Unambiguous operand interpretation in Pass 2

Examples:

(IS, ADD) (R, BREG) (S, ONE)

(DL, DC) (C, 1)

(AD, START) (C, 100)

IC for Statements

Imperative Statement (IS):

```
ADD BREG, ONE
```


IC for Statements

Imperative Statement (IS):

ADD BREG, ONE

(IS, ADD) (R, BREG) (S, ONE)

IC for Statements

Imperative Statement (IS):

ADD BREG, ONE

(IS, ADD) (R, BREG) (S, ONE)

Declarative Statements (DL):

ONE DC 1

IC for Statements

Imperative Statement (IS):

ADD BREG, ONE

(IS, ADD) (R, BREG) (S, ONE)

Declarative Statements (DL):

ONE DC 1

(DL, DC) (C,1)

IC for Statements

Imperative Statement (IS):

ADD BREG, ONE

(IS, ADD) (R, BREG) (S, ONE)

Declarative Statements (DL):

ONE DC 1

(DL, DC) (C,1)

Assembler Directive (AD):

START 1000

END

IC for Statements

Imperative Statement (IS):

ADD BREG, ONE

(IS, ADD) (R, BREG) (S, ONE)

Declarative Statements (DL):

ONE DC 1

(DL, DC) (C, 1)

Assembler Directive (AD):

START 1000

(AD, START) (C, 1000)

END

(AD, END)

Example Pass I

Source Program

```
START 100
MOVER AREG, ='5'
ADD BREG, ONE
ONE DC 1
END
```

Intermediate Code

Location Counter



Symbol Table

Sym_no	Symbol	Address

Literal Table

Lit_no	Literal	Address

Example Pass I

Source Program

```
START 100
MOVER AREG, ='5'
ADD BREG, ONE
ONE DC 1
END
```

Intermediate Code

```
(AD, START) (C,100)
```

Location Counter

100

Symbol Table

Sym_no	Symbol	Address

Literal Table

Lit_no	Literal	Address

Example Pass I

Source Program

```
START 100
MOVER AREG, ='5'
ADD BREG, ONE
ONE DC 1
END
```

Intermediate Code

```
(AD, START) (C,100)
(IS, MOVER) (R,AREG) (L,1)
```

Location Counter

100

Symbol Table

Sym_no	Symbol	Address

Literal Table

Lit_no	Literal	Address
1	='5'	-

Example Pass I

Source Program

START 100
MOVER AREG, ='5'
ADD BREG, ONE
ONE DC 1
END

Intermediate Code

(AD, START) (C,100)
(IS, MOVER) (R,AREG) (L,1)
(IS, ADD) (R, BREG) (S, ONE)

Location Counter

101

Symbol Table

Sym_no	Symbol	Address
1	ONE	-

Literal Table

Lit_no	Literal	Address
1	='5'	-

Example Pass I

Source Program

START 100
MOVER AREG, ='5'
ADD BREG, ONE
ONE DC 1
END

Intermediate Code

(AD, START) (C, 100)
(IS, MOVER) (R, AREG) (L, 1)
(IS, ADD) (R, BREG) (S, ONE)
(DL, DC) (C, 1)
(AD, END)

Location Counter

102

Symbol Table

Sym_no	Symbol	Address
1	ONE	102

Literal Table

Lit_no	Literal	Address
1	='5'	-

Example Pass I

Source Program

```
START 100
MOVER AREG, ='5'
ADD BREG, ONE
ONE DC 1
END
```

Intermediate Code

```
(AD, START) (C, 100)
(IS, MOVER) (R, AREG) (L, 1)
(IS, ADD) (R, BREG) (S, ONE)
(DL, DC) (C, 1)
```

Location Counter

103

Symbol Table

Sym_no	Symbol	Address
1	ONE	102

Literal Table

Lit_no	Literal	Address
1	='5'	103

Example Pass I

Source Program

```
START 100
MOVER AREG, ='5'
ADD BREG, ONE
ONE DC 1
END
```

Intermediate Code

```
(AD, START) (C, 100)
(IS, MOVER) (R, AREG) (L, 1)
(IS, ADD) (R, BREG) (S, ONE)
(DL, DC) (C, 1)
```

Location Counter

103

Symbol Table

Sym_no	Symbol	Address
1	ONE	102

Literal Table

Lit_no	Literal	Address
1	='5'	103

Example Pass I

Source Program

START 100	
MOVER AREG, A	
LOOP: PRINT B	
ADD BREG, ="9"	
SUB BREG, D	
COMP CREG, ="23"	
LTORG	
A DS 5	
LABEL, EQU LOOP	
ORIGIN 800	
L1 MULT CREG, ="7"	
SUB BREG, ="93"	
LTORG	
B DC 10	
MOVEM CREG, ="7"	
PRINT ="7"	
D DC 8	
END	

Symbol Table

Sym.	Symbol	Address

Literal Table

Lit. no	Literal	Address

(MOT)

OP-Code	Mnemonic
01	MOVER
02	MOVEM
03	ADD
04	SUB
05	MULT
06	BV
07	COMP
08	PRINT
09	READ

(POT)

OP-Code	Mnemonic
01	START
02	END
04	ORIGIN
05	LTORG

(DC)

OP-Code	Mnemonic
01	DC
02	DC

REGISTERS

Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Example Pass I

START 100	(AD, 01) - (C, 100)
MOVER AREG, A	100 (15, 01) 01 (S, 01)
LOOP, PRINT B	104 (15, 01) - (S, 01)
ADD BREG, ="9"	102 (15, 04) 02 (L, 1)
SUB BREG, D	106 (15, 04) 03 (L, 05)
COMP CREG, ="23"	104 (AD, 05) - 003
LTORG	107 (AD, 03) - 035
A DS 3	108 (D1, 01) - 03
LABEL, EOU LOOP	
ORIGIN 300	
L. . MULT CREG, ="7"	500 (15, 04) 01 (L, 05)
SUB BREG, ="93"	301 (15, 08) 01 (L, 01)
LTORG	301 (AD, 08) - 501
B DC 10	302 (AD, 09) - 902
MOVEM CREG, ="7"	303 (15, 03) 01 (L, 05)
PRINT ="7"	505 (15, 04) 02 (L, 09)
D DC 8	509 (AD, 02) - 508
END	509 (AD, 02) - 007

Sym.	Symbol	Address
1	A	107
2	LOOP	104
3	B	500
4	D	106
5	LABEL	101
6	L	500

Lit. no	Literal	Address
1	9	105
2	"23"	106
3	"7"	501
4	"93"	502
5	"7"	508

(MOT)	
OP-Code	Mnemonic
01	MOVER
02	MOVEM
03	ADD
05	SUB
06	MULT
06	BV
07	COMP
08	PRINT
09	READ

(POT)	
OP-Code	Mnemonic
01	START
02	END
04	ORIGIN
05	LTORG

(DL)	
OP-Code	Mnemonic
01	DC
02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Intermediate Code

IC does not contain

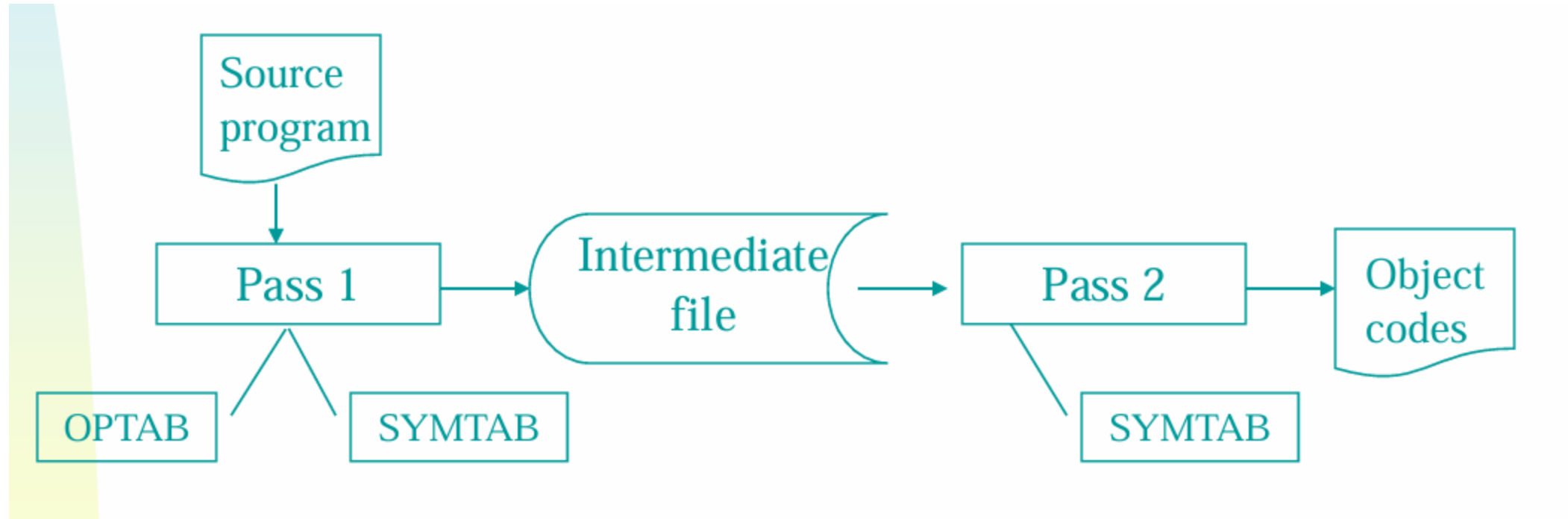
- absolute machine addresses
- binary opcodes
- instruction encoding details

IC is **machine-independent**

Possible Errors in Pass I

- Duplicate Label
- Invalid Operand
- Unrecognized entry in Opcode field

Two Pass Assembler Diagram



Pass II - Synthesis

If no errors are found in the first pass, then the second pass assembles the code into object code.

- Pass 2 follows Pass 1 (Analysis Phase)
- Uses outputs generated in Pass 1
- Responsible for target code generation

Pass 2 does not scan the source program

Objectives

Translate Intermediate Code (IC) into target (object) code

Resolve symbol and literal addresses

Generate final machine-level representation

Pass 2 takes as input:

Intermediate Code (IC)

SYMTAB (Symbol Table)

LITTAB (Literal Table)

OPTAB (Opcode Table)

Role of Instruction Types in Pass II

- **IS (Imperative Statements):** Generate machine instructions
- **DL (Declarative Statements):** Generate data values or reserve space
- **AD (Assembler Directives):** Ignored in Pass 2

Instruction type drives Pass 2 behavior, identifying and replacing with the required machine specific values.

Processing of IS Statements

Imperative Statements (IS) correspond to actual executable machine instructions.

Pass II:

- Identify the opcode field from the IC entry (IS, opcode)
- Fetch the corresponding machine opcode and instruction format from OPTAB
- Decode register operands using the register table (e.g., AREG, BREG)
- Resolve symbol operands using SYMTAB to obtain absolute addresses
- Resolve literal operands using LITTAB to obtain literal addresses
- Assemble the final instruction according to the target machine format

Processing of IS Statements

Source Program

MOVER AREG, ='5'
ADD BREG, ONE

Intermediate Code

(IS, MOVER) (R, AREG) (L, 1)
(IS, ADD) (R, BREG) (S, ONE)

Instruction	Opcode (hex)	Reg	Addr	Final Hex
MOVER AREG, 103	0x04	0x1	0x67	04 1 67
ADD BREG, 102	0x01	0x2	0x66	01 2 66

Processing of DL Statements

Declarative Statements (DL) define or reserve memory; they are not executable instructions.

Pass II :

- Identify the statement type as **DL** from the IC entry (DL, type)
- Distinguish between DC (Define Constant) and DS (Define Storage)
- For DC, extract the constant value from (C, value) in IC
- Convert the constant into target representation (e.g., hex) and place it in memory
- For DS, reserve the specified number of memory locations without initialization
- Update the Location Counter (LC) based on the amount of memory defined or reserved

DL statements affect memory layout but do not generate executable opcodes

Processing of DL Statements

Source Program

ONE DC 1

Intermediate Code

(DL, DC) (C, 1)

Instruction	Opcode (hex)	Reg	Addr	Final Hex
ONE DC 1	-	-	-	01

Processing of AD Statements

Assembler Directives (AD) control the assembly process; they do not represent executable machine instructions.

Pass II :

- Identify the statement type as **AD** from the IC entry (AD, directive)
- Recognize that no opcode lookup in OPTAB is required
- Confirm that all effects of the directive were handled during Pass 1
- Skip machine code generation for the directive
- Maintain correct sequencing in the output (do not disturb LC progression already determined)
- Continue processing the next IC entry until (AD, END) is encountered

No machine code generated in Pass 2, AD statements influence assembly-time behavior only; they never appear in the final machine code output.

Processing of AD Statements

Source Program

START 100

Intermediate Code

(AD, START) - (C, 100)

Instruction	Opcode (hex)	Reg	Addr	Final Hex
START 100	-	-	-	-

Example: Pass I

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

(MOT)	
OP-Code	Mnemonic
01	MOVER
02	MOVEM
03	ADD
05	SUB
06	MULT
06	BV
07	COMP
08	PRINT
09	READ

(POT)	
OP-Code	Mnemonic
01	START
02	END
04	ORIGIN
05	LTORG

(DL)	
OP-Code	Mnemonic
01	DC
02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01)-(C,100)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address

Lit_no	Literal	Address

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP	OP-Code	Mnemonic
08	PRINT	01	DC
09	READ	02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	

Lit_no	Literal	Address

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP	OP-Code	Mnemonic
08	PRINT	01	DC
09	READ	02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	
2	LOOP	101
3	B	

Lit_no	Literal	Address

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,1)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	
2	LOOP	101
3	B	

Lit_no	Literal	Address
1	='9'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,1)
103)(IS,05) 02(S,04)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP	OP-Code	Mnemonic
08	PRINT	01	DC
09	READ	02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	
2	LOOP	101
3	B	
4	D	

Lit_no	Literal	Address
1	= '9'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP	OP-Code	Mnemonic
08	PRINT	01	DC
09	READ	02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	
2	LOOP	101
3	B	
4	D	

Lit_no	Literal	Address
1	= '9'	
2	= '23'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG
B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ	01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	
2	LOOP	101
3	B	
4	D	

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ	01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	
2	LOOP	101
3	B	
4	D	

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ	01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	
4	D	

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	
4	D	
5	LABEL	101

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	
4	D	
5	LABEL	101

Lit_no	Literal	Address
1	='9'	105
2	='23'	106

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ	01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	
4	D	
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	
4	D	
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	
4	= '93'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	
4	D	
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ	01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	
4	D	
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ	01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503
5	= '7'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503
5	= '7'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503
5	= '7'	

Example

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

(MOT)		(POT)	
OP-Code	Mnemonic	OP-Code	Mnemonic
01	MOVER	01	START
02	MOVEM	02	END
03	ADD	04	ORIGIN
05	SUB	05	LTORG
06	MULT	(DL)	
06	BV		
07	COMP		
08	PRINT		
09	READ		
		OP-Code	Mnemonic
		01	DC
		02	DC

REGISTERS	
Reg. No.	Name
01	AREG
02	BREG
03	CREG
04	DREG

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503
5	= '7'	508

Example : Pass II

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

01 - 100

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	='9'	105
2	='23'	106
3	='7'	502
4	='93'	503
5	='7'	508

Example : Pass II

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
```

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	='9'	105
2	='23'	106
3	='7'	502
4	='93'	503
5	='7'	508

Example : Pass II

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
```

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503
5	= '7'	508

Example : Pass II

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
ORIGIN 500
L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
PRINT ='7'
D DC 8
END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
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No IC
500)(IS,06) 03(L,03)
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502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
102)03 03 105
```

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	='9'	105
2	='23'	106
3	='7'	502
4	='93'	503
5	='7'	508

Example : Pass II

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
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LABEL: EQU LOOP
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L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
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END
```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
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503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
102)03 03 105
103)05 02 507
```

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	= '9'	105
2	= '23'	106
3	= '7'	502
4	= '93'	503
5	= '7'	508

Example : Pass II

Source Program

```
START 100
MOVER AREG, A
LOOP: PRINT B
ADD BREG, ='9'
SUB BREG, D
COMP CREG, ='23'
LTORG

A DS 3
LABEL: EQU LOOP
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L1: MULT CREG, ='7'
SUB BREG, ='93'
LTORG

B DC 10
MOVEM CREG, ='7'
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```

Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
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501)(IS,05) 02(L,04)
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503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
102)03 03 105
103)05 02 507
104)07 03 106
```

Sym_no	Symbol	Address
1	A	107
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Example : Pass II

Source Program

```
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MOVER AREG, A
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LABEL: EQU LOOP
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B DC 10
MOVEM CREG, ='7'
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Intermediate Code

```
(AD,01) - (C,100)
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101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
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501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
102)03 03 105
103)05 02 507
104)07 03 106
105)- - 009
```

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1	A	107
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4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
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4	= '93'	503
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Example : Pass II

Source Program

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106)(AD,05) - 023
107)(DL,01) - 03
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No IC
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501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
102)03 03 105
103)05 02 507
104)07 03 106
105)- - 009
106)- - 023
```

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1	A	107
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3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
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Example : Pass II

Source Program

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103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
No IC
500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
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508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
102)03 03 105
103)05 02 507
104)07 03 106
105)- - 009
106)- - 023
107)- - -
```

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
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4	D	507
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6	L1	500

Lit_no	Literal	Address
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Example : Pass II

Source Program

```
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Intermediate Code

```
(AD,01) - (C,100)
100)(IS,01) 01(S,01)
101)(IS,08) - (S,03)
102)(IS,03) 02(L,01)
103)(IS,05) 02(S,04)
104)(IS,07) 03(L,02)
105)(AD,05) - 009
106)(AD,05) - 023
107)(DL,01) - 03
No IC
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500)(IS,06) 03(L,03)
501)(IS,05) 02(L,04)
502)(AD,05) - 007
503)(AD,05) - 093
504)(DL,02) - 010
505)(IS,02) 03(L,05)
506)(IS,08) - (L,05)
507)(DL,02) - (008)
508)(AD,02) - 007
```

Target Code

```
01 - 100
100)01 01 107
101)08 - 504
102)03 03 105
103)05 02 507
104)07 03 106
105)- - 009
106)- - 023
107)- - -
500)06 03 502
501)05 02 503
502)- - 007
503)- - 093
504)- - 010
505)02 03 508
506)08 - 508
507)- - 008
508)- - 007
```

Sym_no	Symbol	Address
1	A	107
2	LOOP	101
3	B	504
4	D	507
5	LABEL	101
6	L1	500

Lit_no	Literal	Address
1	='9'	105
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5	='7'	508